

Respiration

Uses of energy in living organisms:

- Muscle contraction
- Protein synthesis
- Cell division
- Active transport
- Growth
- The passage of nerve impulses
- The maintenance of a constant body temperature.

Aerobic respiration: The chemical reaction in cells that uses oxygen to break down nutrient molecules to release energy.

Glucose + Oxygen $\longrightarrow$ Water + Carbon dioxide



Anaerobic respiration: The chemical reaction in cells that break down nutrient molecules to release energy without using oxygen.

* Releases much less energy per glucose molecule than aerobic respiration.

- In yeast: Glucose $\longrightarrow$ Alcohol + Carbon dioxide



- In muscles: Glucose $\longrightarrow$ lactic acid

The effect of temperature in yeast production:

- Yeast respire anaerobically.
- Yeast requires sugar to be activated in order to make anaerobic respiration
- Carbon dioxide produced will make the dough rise
- The process is controlled by enzymes. Enzymes work best from $35^\circ - 40^\circ C$, so you can increase the temperature up to $40^\circ C$ to increase the respiration of the yeast
- If you increase the temperature higher than $40^\circ C$, the enzyme becomes denatural and respiration decreases

Oxygen debt:

- At one point during exercise, oxygen does not reach the muscles fast enough, so muscles respire anaerobically, which releases lactic acid.
- Lactic acid builds up in muscles and blood during vigorous exercise, causing muscle cramps and oxygen debt.
- The lactic acid produced is toxic and a poison to muscles as it causes muscle cramps and aches
- Cells get rid of lactic acid by converting it into carbon dioxide and water. Cells need oxygen to carry out this process
- This extra oxygen is called oxygen debt

In liver cells: lactic acid + Oxygen $\longrightarrow$ Carbon dioxide + Water

How is oxygen debt removed?

- Continuation of fast heart rate to transport lactic acid in the blood from the muscles to the liver.
- Continuation of deeper and faster breathing to supply oxygen for aerobic respiration of lactic acid
- Aerobic respiration of lactic acid in the liver.